

distributional lift
 $J, K, V \rightarrow (\tau, \mathcal{K})$

kernel handoff
 $I_{\text{fix}}, K \rightarrow S, g$

the single-cell process closed single-cell theory

fixation functions
 $I(t), D(t), I_{\text{fix}}(t)$

closed moments
and release
 $J(t), K(t), V(t)$

proved identities
 $I'(t) = -\delta J(t),$
 $V'(t) = \delta K(t)$

its distribution theory distributional laws and MOI

joint event description
burst time τ
and burst size $\mathcal{K}$

geometric burst
law and QSD
 $\Pr\{\mathcal{K} = k\} = (\delta/\lambda) a^{-k}$

multiple founders
 $I^k, K_k,$ and
 $g_k(t) = \delta K_k(t)$

this chapter burst-aware population renewal

mechanistic kernels
 $S(\alpha) = I_{\text{fix}}(\alpha),$
 $g(\alpha) = \delta K(\alpha)$

cohort sums
 $\mathcal{I} = \mathcal{I}_0 S + i * S$
 $\mathcal{V}' = \mathcal{I}_0 g + i * g - c\mathcal{V}$

population feedback
 $i(t) = \gamma T \mathcal{V}(t)$

closed formulae replace simulation-only inputs timing, size, and MOI are retained as laws cell-age structure enters incidence dynamics

Closed moments $\rightarrow$ laws of $(\tau, \mathcal{K})$ and MOI $\rightarrow$ renewal kernels $\rightarrow$ population feedback